



Efficiently removal of ciprofloxacin from aqueous solution by MIL-101(Cr)-HSO₃: the enhanced electrostatic interaction

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Abstract

Metal–organic frameworks (MOFs) have been widely used to remove organic/toxic compounds from waste water. Ciprofloxacin (CIP) has been detected in surface and waste water, which is harmful to aquatic organisms and human body. Herein, MIL-101(Cr)-HSO₃ was synthesized by solvothermal method and its structural features were characterized by XRD, SEM, FTIR, N₂ adsorption–desorption analysis at 77 K and zeta potential. Then, the CIP adsorption performance of MIL-101(Cr)-HSO₃ was investigated, in which the effect of adsorbent dosage, contact time, pH and ionic strength were explored. MIL-101(Cr)-HSO₃ showed the highest adsorption capacity when the adsorbent dosage was 0.1 g/L and the pH was 8.0. The observation from the effects of pH and ionic strength suggested a stronger electrostatic interactions between CIP and MIL-101(Cr)-HSO₃. The pseudo-second-order model fitted the adsorption kinetics data of MIL-101(Cr)-HSO₃ well. Moreover, the equilibrium adsorption data of MIL-101(Cr)-HSO₃ followed the Langmuir model, indicating a mono-layer adsorption of CIP onto surface of MIL-101(Cr)-HSO₃. The calculated maximum CIP adsorption capacity from Langmuir model was 564.9 mg/g, which was higher than the reported materials. Besides, the equilibrium adsorption data were fitted to the Tempkin model with $r^2 = 0.9880$, which also suggested a stronger electrostatic interaction between CIP and MIL-101(Cr)-HSO₃. Finally, the introduced sulfonic acid group made the material more negatively charged on the surface, which benefited the adoption of CIP via stronger electrostatic interactions resulting the enhanced adsorption capacity of CIP. The results show that the MIL-101(Cr)-HSO₃ is a promising candidate for removal of CIP and introducing proper functional groups on organic linker is a convenient way to obtain MOFs with better performance for a specific application.

Keywords Metal–organic frameworks · Sulfonic acid group · Ciprofloxacin · Adsorption · Electrostatic interaction

1 Introduction

Fluoroquinolone antibiotics (FQAs) are wide spectrum antibacterial and have been extensively used in human therapy, cattle industry, and aquaculture [1, 2]. Among

the FQAs, ciprofloxacin (CIP, C₁₇H₁₈FN₃O₃) is one of the most widely used fluoroquinolone antibiotics in the world and its molecular formula is illustrated in Fig. S1 [3–5]. Since the low biodegradability of CIP, the traditional biochemical treatment method cannot completely remove the residuals of CIP from waste water [6, 7]. Hence, CIP has been detected in surface water and waste water. The CIP concentrations in surface water and waste water reported in the previous works are µg/L and mg/L, respectively [5, 8–10]. However, the residuals of CIP remaining in water is harmful to some aquatic organisms even at low concentrations. Moreover, the presence of CIP in environment also can fuel an increase in antibiotic-resistant bacteria, which would arouse apparent potential threat to human health [2, 3, 8, 11, 12]. Therefore, it is particularly urgent and important to develop efficient treatment methods to eliminate CIP from waste water. Consequently, various technologies and methods have been developed to remove

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CIP, including oxidation [13–16], photo degradation [17], membrane technology [18] and adsorption method [6, 7, 19–21]. Compared with the other technologies, adsorption has been regarded as one of the most promising ones because of its simple equipments, easy operation, high efficiency and economic benefits [22, 23]. Also, adsorption technology does not produce toxic intermediates [24]. Hence, the adsorption performance of several kinds of porous adsorbents for CIP removal from aqueous solutions have been explored, such as activated carbon [19, 25, 26], montmorillonite [27, 28], carbon nanotubes [29, 30], biochars [31, 32], and zeolites [33]. It can be concluded that most of the porous materials showed relatively low adsorption capacity, which would hinder their application in CIP removal. Besides, the key of the adsorption technology is to find/design materials with better adsorption performance of CIP.

Recently, metal–organic frameworks (MOFs), one of the novel functional porous materials, is built by connecting various transition metal ions with a variety of organic linkers [34]. MOFs has attracted an increasing research interests owing to its unique structural characteristics, such as large surface area, high porosity, tunable pore size and surface functionalities. Hence, MOFs has been considered as a promising potential porous material in several applications, including gas storage [35], gas/liquid adsorption and separation [36–39], catalysis [40, 41] and drug delivery [42]. It should be noticed that a variety of hydrothermally stable MOFs have been used to remove organic/toxic component and metal ions from liquid phase [38, 39, 43, 44]. Especially, the CIP adsorption performance of several MOFs, magnetic MOFs and MOF-derived porous materials were determined in the previous works. Bayazit et al. prepared the MIL-101(Cr)/Fe₃O₄ composite with the maximum adsorption uptake of 63.28 mg/g at 298 K, while the maximum adsorption capacity of the Fe₃O₄/Cu-BTC and Fe₃O₄/MIL-100(Fe) composite were 538 mg/g and 323 mg/g, respectively [23, 24, 45]. Besides, several MOF-derived porous materials were used as efficient adsorbents for CIP removal. The ZIF-8-derived nanoporous carbons showed good recyclability and a high adsorption capacity of 417 mg/g [46], whilst the adsorption capacity of the silica coated ZIF-8 derived C@silica core/shell nanoparticles reached a high adsorption capacity of 517 mg/g [7]. Liang et al. developed an in situ synthetic method to prepare hollow Co₃S₄ by ZIF-67 and the hollow Co₃S₄ can be regarded as a superior adsorbent with a high CIP adsorption capacity of 472 mg/g [47]. The above adsorption capacity of MOF-derived porous materials was higher than other reported porous materials, indicating that MOFs can serve as a promising material alternative for CIP removal. However, the CIP adsorption capacity of these MOF-derived materials needs to be further enhanced and the synthesis process was relatively complicated.

It has been proved that electrostatic interaction and π – π interaction affect the adsorption performance of CIP onto porous materials [20, 21, 24, 29, 30]. At the same time, the CIP molecules exist as cationic species at pH lower than 5.90, as zwitterionic species at pH ranging from 5.90 to 8.89, and as anionic species at pH higher than 8.89 (see Figs. S1 and S2) [26]. Hence, if we can change the surface charges of selected materials to match the existing formation of CIP at different pH, for example, the surface charges of porous materials were negative at pH lower than 5.90, the adsorption capacity of the selected materials may be higher due to the enhanced electrostatic interactions. Among the various MOFs with good hydrothermal stability, MIL-101(Cr) and its derivatives have been widely used to treat wastewater and remove sulfide and nitride compounds from fuel oil [48–52]. MIL-101(Cr) and its derivatives showed good adsorption performance for the above mentioned species because of its good hydrothermal stability and high surface area. However, the surface of MIL-101(Cr) was charged positively at pH lower than 7.0 resulting in a lower CIP adsorption capacity [45]. It has been found that introducing functional groups on organic linkers is a feasible way to change the surface charge of MOFs [53]. To change the surface charges of MIL-101(Cr), the electron withdrawing group can be added on the organic linker to make the surface negatively charged, such as –NO₂, –CN, –HSO₃, and Br. Besides, the inductive effect of –HSO₃ is in the middle compared with the other electron withdrawing groups. Hence, the terephthalic acid was functionalized by –HSO₃ to prepare MIL-101(Cr)-HSO₃, which may possesses negative surface charges matching the formation species of CIP at different values of pH and show higher CIP adsorption capacity. What's more, the relatively simple preparation process also promoted us to choose MIL-101(Cr)-HSO₃. In this work, MIL-101(Cr)-HSO₃ was facilely prepared through a simple solvothermal method and the structural features of the as-synthesized MIL-101(Cr)-HSO₃ was characterized. Then, its CIP adsorption performance was explored thoroughly. To obtain the optimal adsorption parameters for CIP, various experimental factors were evaluated in detail, including adsorbent dosage, solution pH, contact time and ionic strength. Additionally, the adsorption kinetics, adsorption isotherms, and thermodynamic behaviors of MIL-101(Cr)-HSO₃ were investigated. At last, the adsorption mechanism of CIP on MIL-101(Cr)-HSO₃ was also deuced.

2 Materials and methods

2.1 Materials

CIP (98%) and terephthalic acid (BDC) were purchased from Macklin Biochemical Co., Ltd (Shanghai, China). CrO₃ and

$\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ were obtained from Xilong Scientific Co., Ltd (Shantou, China). Monosodium 2-sulfoterephthalic acid ($\text{BDC-SO}_3\text{Na}$) was purchased from TCI Development Co., Ltd (Shanghai, China). Methanol, ethanol and acetone were obtained from Tianjin Yongda Chemical Reagent Co., Ltd. *N,N*-dimethylformamide (DMF, 99.5%) was purchased from Tianjin Jindongtianzheng Precision Chemical Reagent Factory. The other reagents, such as NaOH, HCl, HF, and NaCl were obtained from Beijing HWRK Chemical Co., Ltd. The purchased chemicals were all analytical grade and used as received without further purification. Deionized water was used as solvent to prepare all of the solutions.

2.2 Synthesis of MIL-101 and MIL-101- HSO_3

MIL-101(Cr) and MIL-101(Cr)- HSO_3 were synthesized by using the hydrothermal method described in the literature with some modifications [54, 55]. MIL-101(Cr) was prepared following the procedure taken from Férey's paper: 2.0 g $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 0.82 g H_2BDC were dissolved in 24 mL deionized water with 0.2 mL HF. The mixture was introduced into 100 mL Teflon-lined vessel, which was placed into a preheated oven and held at 493 K for 8 h. After cooling down, the green powder containing recrystallized H_2BDC was separated by filtration. Then, the product was washed by water and DMF for three times, respectively. The green powder was dried in a vacuum oven at 353 K for 12 h. To eliminate the residual H_2BDC molecules in the pores, the dried product was further treated by the following process. The activation treatment was performed using two subsequent solvent treatments, DMF at 333 K for 6 h and hot ethanol at 393 K for 24 h in a Teflon-lined stainless steel autoclave. At last, the precipitate was filtered and washed with 100 mL acetone, and dried overnight at 393 K in a vacuum oven. MIL-101(Cr)- HSO_3 was prepared using the following process. 3.35 g of monosodium 2-sulfoterephthalic acid, 1.25 g of CrO_3 and 0.91 g of concentrated aqueous hydrochloric acid (12 mol/L) were dissolved in 50 mL water. Then this highly acidic solution was transferred to a Teflon-lined vessel and heated at 453 K for 6 days. After cooling naturally, the reaction solid product was obtained by filtration and washed three times with deionized water (400 mL) and methanol (100 mL). At last, the product was dried overnight at 353 K in a vacuum oven. The synthesis process for the other water-stable MOFs was listed in the supplementary data.

2.3 Characterization of synthesized MOFs

The powder X-ray diffraction patterns (XRD) of the selected MOFs were obtained by using a D/MAX-2500 diffractometer (Rigaku, Japan) with Cu K α radiation (40 kV, 150 mA) and the range of the 2θ degree was from 1° to 50° . The N_2

adsorption–desorption isotherms of MOFs were measured at 77 K with a surface area analyzer (Autosorb-IQ-MP, Quantachrome, America). Based on the N_2 adsorption–desorption isotherm at 77 K, the specific surface area and pore size distribution were calculated by BET and BJH method, respectively. The PHS-3C pH meter (Shanghai INESA Scientific Instrument Co., Ltd., China) was used to test the pH value of the prepared solutions. The zeta potentials of MOFs were obtained by a nano-ZS90 Zeta potential analyzer (Malvern Instruments, England) to determine the surface charges of MOFs. X-ray photoelectron spectroscopy (XPS) measurements were conducted on a Kratos Axis Ultra DLD XPS system equipped with an Al X-ray source. The transmission electron microscope (TEM) was utilized to observe the internal structure of MIL-101(Cr)- HSO_3 and regenerated MIL-101(Cr)- HSO_3 using lanthanum hexaboride filament as electron emission source (Japan, JEOL, JEM-2100, 80–200 kV).

2.4 Adsorption experiments

All adsorption experiments were performed in series of 250 mL conical flasks, which were placed in a thermostatic shaker (THZ-92B, Shanghai Boxun Medical Biological Corp., Shanghai, China) at 300 rpm and different temperatures. Each flask contained 100 mL CIP solution and a certain amount of adsorbents. The effects of adsorbent dosage, pH value, and ionic strength on the adsorption performance of CIP were investigated for MIL-101(Cr)- HSO_3 . For the adsorbent dosage, a series of the adsorbent (5, 10, 20, 50, 100, 200 and 300 mg) was added into 100 mL CIP solution with a concentration of 120 mg/L, respectively. For the evaluation of pH effect, the adsorption experiments were performed by adding 10 mg MIL-101(Cr)- HSO_3 into 100 mL CIP solution with designated pH values and the concentration of CIP solution was also 120 mg/L. The pH value of solutions ranged from 2 to 12, which was adjusted by 0.1 M HCl or 0.1 M NaOH aqueous solution. To examine the effect of ionic strength, predetermined amounts of NaCl and 10 mg MIL-101(Cr)- HSO_3 were added into the flasks with 100 mL of 120 mg/L CIP solution to obtain 0.01 M, 0.1 M, 0.5 M, and 1.0 M ionic strength solutions. Then, all the prepared flasks were shaken in a thermostatic shaker at predetermined temperature for 24 h to guarantee adsorption equilibrium. After adsorption, the samples were filtered by a 10 mL syringe and filter membrane with the pore size of 0.22 μm . The obtained solution was diluted with deionized water and its concentration of CIP was measured by ultraviolet–visible (UV–Vis) spectrophotometer (TU-1901, Beijing Purkinje General Instrument CO., Ltd, Beijing, China) at 272 nm. Besides, the adsorption capacity of several other water-stable MOFs were also explored in flasks with 100 mL of 120 mg/L CIP solution and 10 mg adsorbent at 303 K.

To obtain the adsorption isotherms of MIL-101(Cr)-HSO₃, adsorption experiments were conducted with 100 mL CIP solutions with different concentrations (15–120 mg/L) and 10 mg adsorbent at pH 8.0. The samples were shaken at a speed of 300 rpm and different temperatures for 24 h to guarantee adsorption equilibrium. After adsorption, the residual concentration of CIP was determined by UV–Vis spectrophotometer at 272 nm. In the study of adsorption kinetics, 10 mg MIL-101(Cr)-HSO₃ was added into each flask containing 100 mL of 120 mg/L CIP solution at pH 8.0. Then the flasks were putted into a thermostatic shaker and shaken at 300 rpm and 303 K for different time (from 0 to 24 h). To avoid the possible interactions, one flask represented one adsorption time. At predetermined time intervals, the samples were filtered by 0.22 µm membrane and the CIP concentration was analyzed by UV–Vis spectrophotometer.

The adsorption amount of CIP (q , mg/g) was calculated by the following equation (Eq. 1):

$$q = \frac{(c_0 - c_e)V}{m_{\text{adsorbent}}} \quad (1)$$

where V (L) is the volume of CIP solution used in the adsorption process, c_0 and c_e (mg/L) are the initial and equilibrium concentration of CIP, and $m_{\text{adsorbent}}$ is the mass of the responding adsorbent. In this study, all the adsorption experiments were conducted in duplicate. The final adsorption capacity of adsorbent was taken from the mean values. Besides, blank experiments were also performed to eliminate the influence of flasks on the CIP adsorption capacity.

3 Results and discussion

3.1 Characterization of synthesized MOFs

MIL-101(Cr) and MIL-101(Cr)-HSO₃ were prepared by thermal method and characterized by XRD, N₂ adsorption–desorption isotherms, FTIR and SEM as discussed below. The XRD patterns of MIL-101(Cr) and MIL-101(Cr)-HSO₃ were illustrated in Fig. 1, while the XRD patterns of other MOFs were shown in Fig. S3. The peaks of the synthesized MIL-101(Cr) were consistent with that of the simulated MIL-101(Cr), indicating that the obtained materials were confirmed to be MIL-101(Cr). Besides, the similarity of the XRD patterns between MIL-101(Cr) and MIL-101(Cr)-HSO₃ indicated that the introduced –HSO₃ group did not change the crystal structures of MIL-101(Cr). Figure 2a showed the nitrogen adsorption/desorption isotherms of MIL-101(Cr)-HSO₃ at 77 K and the pore size distribution analysis. The BET surface area of MIL-101(Cr)-HSO₃ were 1545.4 m²/g, which was close to the reported value of MIL-101(Cr)-HSO₃ (around 1580 m²/g) [56–58] (Fig. 2).

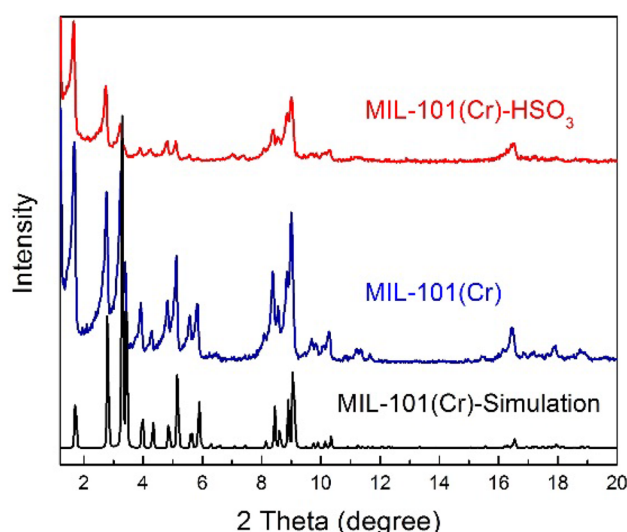


Fig. 1 XRD patterns of MIL-101(Cr) and MIL-101(Cr)-HSO₃

The pore size of MIL-101(Cr)-HSO₃ was about 3.07 nm, which was a little larger than the reported values. The FTIR spectra of MIL-101(Cr)-HSO₃, CIP and MIL-101(Cr)-HSO₃ adsorbed CIP were shown in Fig. 3. For MIL-101(Cr)-HSO₃ (see Fig. 3a), the bands at 1402 and 1492 cm^{−1} can be assigned to the characteristic vibration of the dicarboxylate groups (O–C–O)– [45]. The specific sulfonic acid stretching appears between the wavenumbers 1800 and 1000 cm^{−1}. The bands of O=S=O symmetric stretching, the in-plane skeletal vibration of the benzene rings substituted by a sulfonic acid group, and the S–O stretching vibration can be found at 1179, 1080, and 1025 cm^{−1}, respectively [59, 60]. Besides, the band observed at 670 cm^{−1} may be attributed to C–S stretching vibration [61]. Hence, the –HSO₃ functional group was introduced on MIL-101(Cr)-HSO₃, which was confirmed by the FTIR spectrum. For CIP (see Fig. 3b), the band appearing at 1700 cm^{−1} may correspond to the vibrations of carboxylic acid group C=O, whereas the band observed at 1628 cm^{−1} may be attributed to the vibrations of ketone group. The band at 1383 cm^{−1} indicated the vibration of the protonation of amine group in the piperazine moiety, while the band at 1270 cm^{−1} can be assigned to coupling of vibrations of the carboxyl and OH deformation [26, 62]. It can be seen that the FTIR spectrum of MIL-101(Cr)-HSO₃ had changed a little after CIP adsorption, as shown in Fig. 3c. Compared with the spectra of CIP, the absence of 1700 cm^{−1} may be caused by the deprotonation of the carboxyl group. The peak observed at 1550 cm^{−1} in CIP and MIL-101(Cr)-HSO₃ adsorbed CIP samples also indicated that the carboxyl group can be regarded as COO[−] group [63, 64]. The band appeared at 1270 cm^{−1} indicating the coupling of vibrations of the carboxyl and OH deformation can be found in MIL-101(Cr)-HSO₃ adsorbed CIP sample.

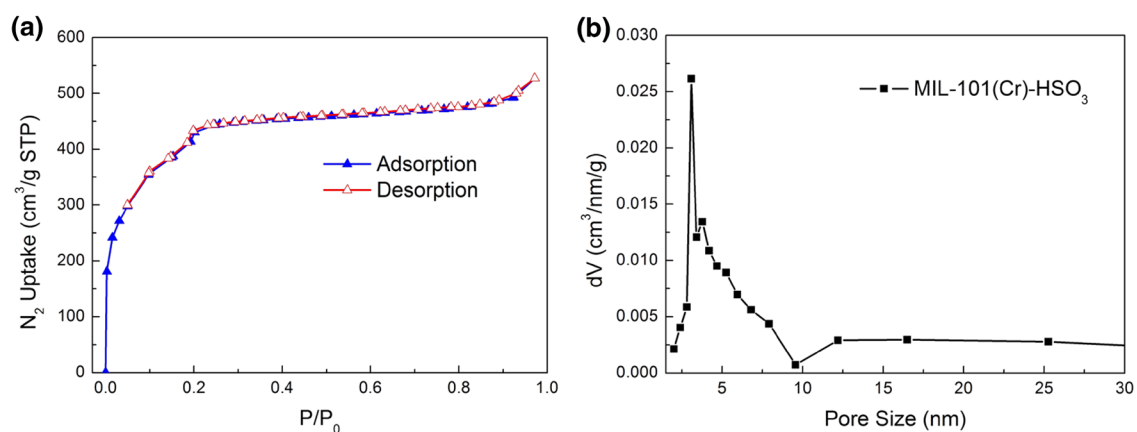


Fig. 2 **a** N_2 adsorption-desorption isotherms for MIL-101(Cr)-HSO₃ at 77 K, **b** pore size distribution analysis for MIL-101(Cr)-HSO₃

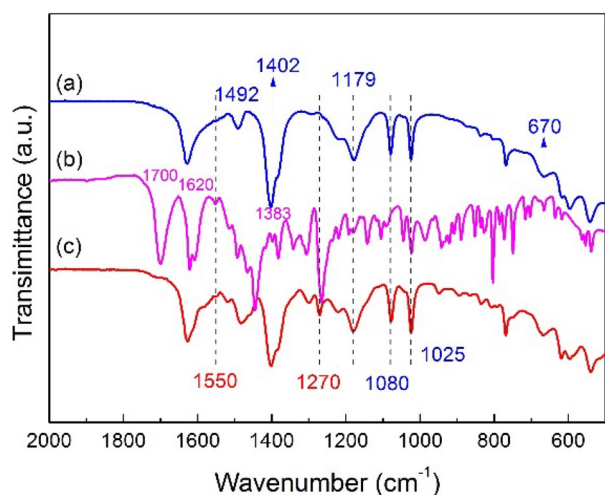


Fig. 3 FTIR spectra of (a) MIL-101(Cr)-HSO₃, (b) CIP and (c) MIL-101(Cr)-HSO₃ adsorbed CIP in range 500–2000 cm^{-1}

What's more, the bands at 745 cm^{-1} , 948 cm^{-1} , 1301 cm^{-1} and 1518 cm^{-1} that were assigned to the functional groups of CIP were also observed in the MIL-101(Cr)-HSO₃ adsorbed CIP sample, indicating that CIP was adsorbed on the surface of MIL-101(Cr)-HSO₃ successfully. And, the band at 1270 cm^{-1} did not shift after CIP adsorption, suggesting that the interaction between MIL-101(Cr)-HSO₃ and carboxyl group existed [62]. The particle size and morphology of the synthesized MIL-101(Cr)-HSO₃ samples were measured by SEM, as shown in Fig. 4. The SEM images of MIL-101(Cr)-HSO₃ showed a good crystallinity and the crystal size of the octahedral particles was ranged from 200 to 500 nm.

3.2 Effect of MIL-101-HSO₃ dosage

In order to investigate the influence of -HSO₃ group on adsorption performance of MIL-101(Cr), adsorption experiments were conducted by using MIL-101(Cr) and

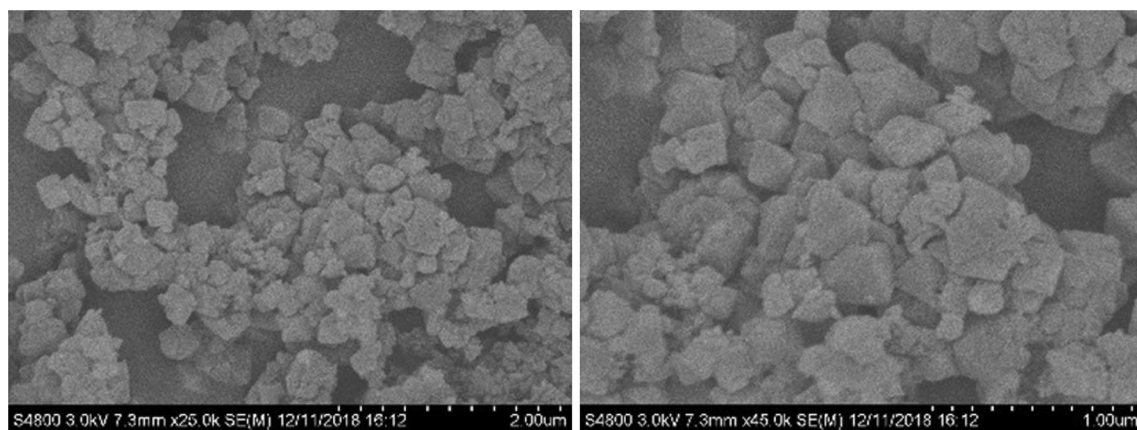


Fig. 4 The SEM images of MIL-101(Cr)-HSO₃

MIL-101(Cr)-HSO₃ as adsorbent. Besides, the CIP adsorption performance of several other water-stable MOFs was also studied. The results were shown in Fig. S4. The CIP adsorption capacity of MIL-101(Cr)-HSO₃ was 453.11 mg/g, which was around 4.0 times higher than that of MIL-101(Cr) and also higher than the other selected MOFs. This significant improvement can be attributed to the introduced -HSO₃ group. Hence, the factors, like pH, content time, and dosage, which influenced its adsorption performance of CIP, were explored by the following experiments.

Firstly, the influence of adsorbent dosage was explored by varying the MIL-101(Cr)-HSO₃ amount from 0.05 to 3.0 g/L as shown in Fig. 5. The adsorption capacity of CIP increased from 431.55 to 469.50 mg/g, while the dosage of MIL-101(Cr)-HSO₃ increased from 0.05 to 0.1 g/L. Then, the adsorption capacity of CIP declined from 469.50 to 38.99 mg/g, when the dosage increased from 0.1 to 3.0 g/L. At lower dosage of adsorbent, the adsorption sites were exposed and can be saturated by CIP faster, indicating a higher adsorption capacity. When the dosage increased, the excess number of unoccupied adsorption sites caused a lower adsorption capacity [6, 47]. Hence, the MIL-101(Cr)-HSO₃ dosage of 0.1 g/L was selected for the further adsorption experiments.

3.3 Effect of pH and contact time

The existing form of adsorbate and surface adsorption sites of adsorbent are effectively determined by the pH value of solution, which has significant influence on the adsorption process. Therefore, adsorption experiments were conducted to investigate the effects of pH. The pH value of solutions ranged from 2.0 to 12.0. As shown in Fig. 6, the adsorption capacity of MIL-101(Cr)-HSO₃ for CIP increased with the

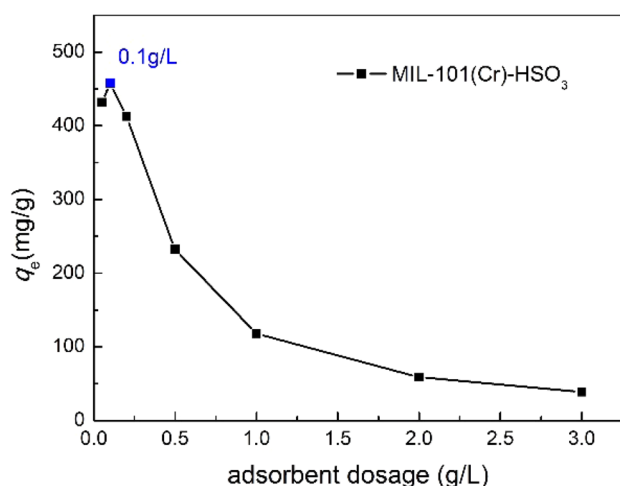


Fig. 5 Effect of adsorbent dosage on adsorption of CIP on MIL-101(Cr)-HSO₃ (T = 303 K, contact time = 24 h, c_0 = 120 mg/L)

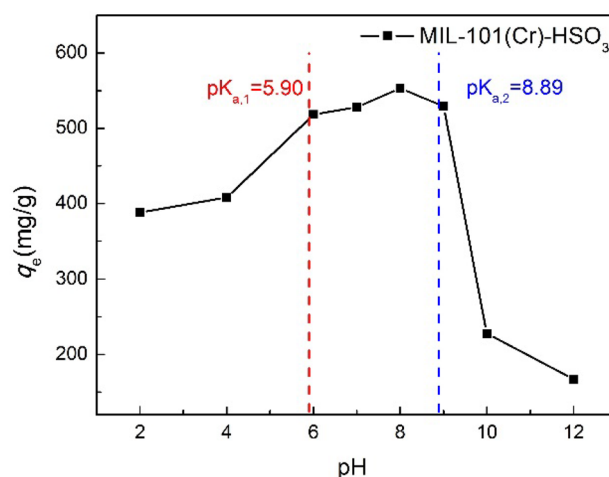


Fig. 6 Effect of pH on adsorption performance of MIL-101(Cr)-HSO₃ for CIP (adsorbent dosage = 0.1 g/L, T = 303 K, contact time = 24 h, c_0 = 120 mg/L)

solution pH increasing from 2.0 to 8.0. Then, the adsorption capacity decreased with the increasing pH from 8.0 to 12.0. As shown in Figs. S1 and S2, CIP has two pK_a values of 5.90 and 8.89, indicating that the groups of CIP can form three species by protonation–deprotonation reaction in aqueous solutions with different pH values. Also, CIP molecules existed as cationic and zwitterionic species when $pH < 5.90$ and the pH value ranged from 5.90 to 8.89. When the solution pH was higher than 8.89, the main formation of CIP was anionic species [26]. The zeta potentials of MIL-101(Cr) (Fig. 7a) at different pH values varying from 3.0 to 7.0 suggested that the surface of MIL-101(Cr) was positively charged. This suggested the reason of lower adsorption capacity of MIL-101(Cr): the initial pH of CIP solution with concentration of 120 mg/L was around 5.0 suggesting that the CIP existed as cations and the surface of MIL-101(Cr) was also kept as positive charged one. Hence, the electrostatic repulsion between CIP cations and positive charged surface of MIL-101(Cr) resulted in its lower adsorption capacity. As shown in Fig. 7b, it is clear that the surface of MIL-101(Cr)-HSO₃ was negatively charged in the whole tested pH range, which validated the hypothesis brought in the introduction section. Hence, unlike MIL-101(Cr), the electrostatic attraction between the negative charged surface of MIL-101(Cr)-HSO₃ and CIP cations benefited its adsorption capacity at pH 5.0. What's more, the zeta potentials of MIL-101(Cr) obtained from pH 7.0 to 8.89 were lower than that of MIL-101(Cr)-HSO₃, which can speculate that the adsorption capacity of MIL-101(Cr)-HSO₃ would be higher than that of MIL-101(Cr). Interestingly, the zeta potentials of MIL-101(Cr)-HSO₃ were decreased sharply when the solution pH ranged from 3.0 to 6.0, while minor changes were observed in the pH range of 6.0–10.0. Considering the

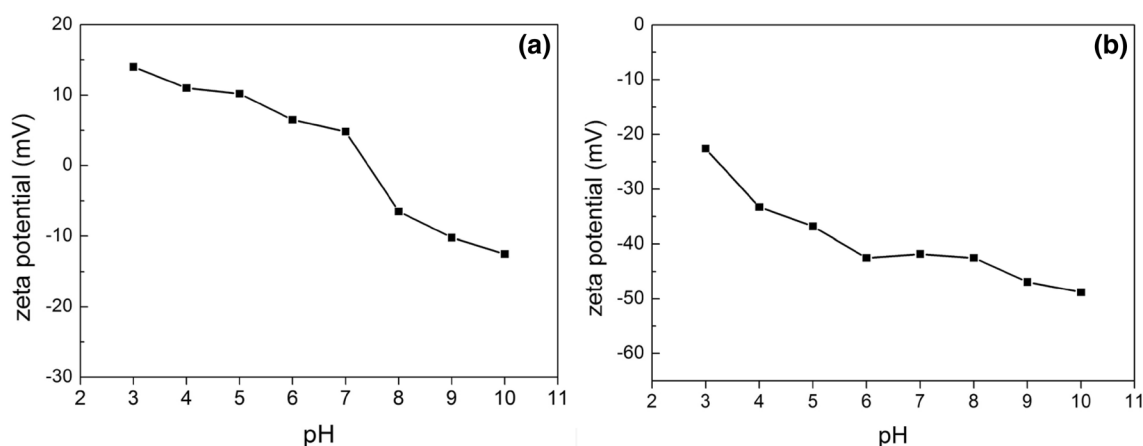


Fig. 7 Zeta potentials of **a** MIL-101(Cr) and **b** MIL-101(Cr)-HSO₃ at different pH values

charged state of CIP and MIL-101(Cr)-HSO₃ at different pH values, the electrostatic attraction and hydrophobic interaction may play important roles during the adsorption process [24, 29, 46, 47]. For pH < 6.0, the less electrostatic attraction between cationic species of CIP and less negatively charged surfaces of MIL-101(Cr)-HSO₃ and the competition between H⁺ and cationic species of CIP may cause the relatively lower adsorption capacity. Besides, the -NH of CIP was protonated to NH₂⁺ that led to the disappearance of hydrogen bonding with the O atom of Cr-O-C in MIL-101(Cr)-HSO₃ was the second reason for the relatively low adsorption capacity. At pH > 9.0, the anionic species of CIP was increased gradually and the surface negative charge of MIL-101(Cr)-HSO₃ was also enhanced, which enhanced the electrostatic repulsion resulting in the decrease of adsorption capacity. In the pH values ranged from 6.0 to 8.0, the surface negative charge of MIL-101(Cr)-HSO₃ was not changed obviously and CIP mainly existed as zwitterions with a fraction of cations/anions, in which the anionic species of CIP increased with the increasing pH values. The electrostatic attraction between NH₂⁺ and negative charge of MIL-101(Cr)-HSO₃ and the π - π interaction between the benzene rings within MIL-101(Cr)-HSO₃ and CIP played important roles during the adsorption process, resulting in the relatively high adsorption capacity. Besides, the COO⁻ groups in zwitterionic species of CIP can coordinate with the open metal sites in MIL-101(Cr)-HSO₃, which may cause the high adsorption capacity. Hence, MIL-101(Cr)-HSO₃ showed the highest adsorption capacity at pH 8.0. Similar results were also observed for CIP adsorption by Fe₃O₄/HKUST-1, MIL-101/Fe₃O₄ and montmorillonite [43, 64]. Therefore, pH 8.0 was selected as the optimal pH for further investigation of CIP adsorption by MIL-101(Cr)-HSO₃.

The adsorption rate is another important factor affecting the CIP removal process from aqueous solutions. Figure 8 showed the effect of contact time on the CIP adsorption

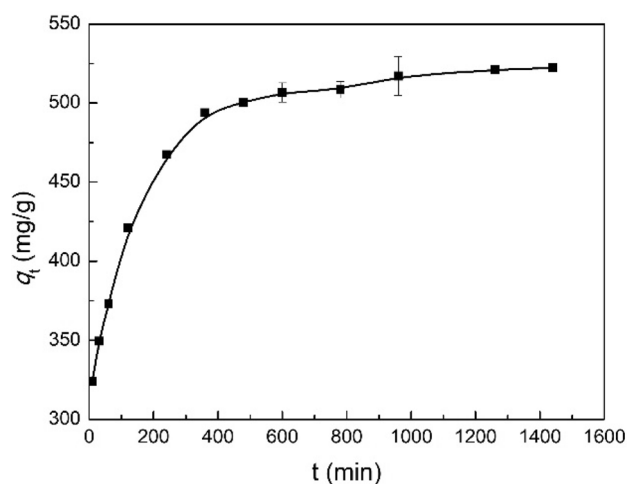


Fig. 8 Effect of contact time on the adsorption capacity of CIP by MIL-101(Cr)-HSO₃ (adsorbent dosage = 0.1 g/L, T = 303 K, pH 8.0, c₀ = 120 mg/L)

by MIL-101(Cr)-HSO₃. It was evident that the adsorption of CIP onto MIL-101(Cr)-HSO₃ was relatively fast in the first 360 min. The fast adsorption rate at the beginning may be attributed to the high CIP concentration gradient between CIP solution and MIL-101(Cr)-HSO₃ as well as the vacancy of the adsorption sites of MIL-101(Cr)-HSO₃ adsorbent. Then, the adsorption rate decreased with further increasing the contact time and adsorption reached the equilibrium within 24 h. When the adsorption sites were occupied by CIP gradually, the diffusion rate encountered great resistance resulting in that a minor change was observed on adsorption capacity. Hence, to guarantee the adsorption equilibrium of MIL-101(Cr)-HSO₃, 24 h was selected as the optimal contact time for the subsequent experiments.

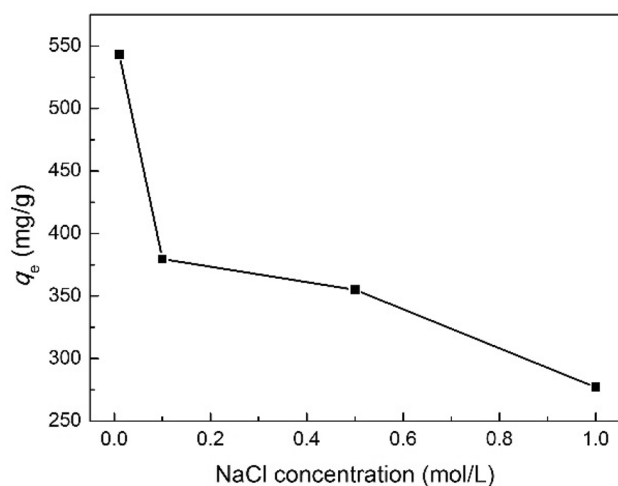


Fig. 9 Effect of ionic strength on the adsorption of CIP by MIL-101(Cr)-HSO₃ (adsorbent dosage=0.1 g/L, T=303 K, pH 8.0, contact time=24 h, c_0 =120 mg/L)

3.4 Effect of ionic strength

The experiments indicating the influence of ionic strength on the CIP adsorption by MIL-101(Cr)-HSO₃ were conducted in NaCl solution with concentration ranging from 0.01 to 1.0 mol/L, as shown in Fig. 9. It was found that the adsorption capacity suddenly decreased when the NaCl concentration increased to 0.1 mol/L. This result can be attributed to the following two reasons. Firstly, the viscosity and density

of the solution increased with the increasing NaCl concentration, which hindered the mass transfer of CIP to the solid [24]. Secondly, the electrostatic attraction between CIP⁺ and adsorbent was weakened owing to that Na⁺ can neutralized the surface charge of MIL-101(Cr)-HSO₃, which led to the decrease of adsorption capacity. Therefore, the subsequent investigations were carried out without NaCl addition.

3.5 Adsorption kinetics

To estimate the rate of adsorption process on CIP, pseudo-first-order kinetic model and pseudo-second-order kinetic model were used to fit the adsorption kinetic data. Both the equations were represented as follows:

$$\text{Pseudo-first-order kinetic model: } \ln(q_e - q_t) = \ln q_e - k_1 t \quad (2)$$

$$\text{Pseudo-second-order kinetic model: } \frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (3)$$

where t (min) is the adsorption time, q_e (mg/g) is the equilibrium adsorption capacity, q_t (mg/g) is the adsorption capacity at time t , k_1 (min⁻¹) and k_2 (g/(mg·min)) are the adsorption rate constants for the pseudo-first-order and pseudo-second-order model, respectively. The fitted plots of two kinetic models were shown in Fig. 10 and the values of the kinetic parameters of two models were displayed in Table 1. The correlation coefficient value (R^2) of the pseudo-first-order

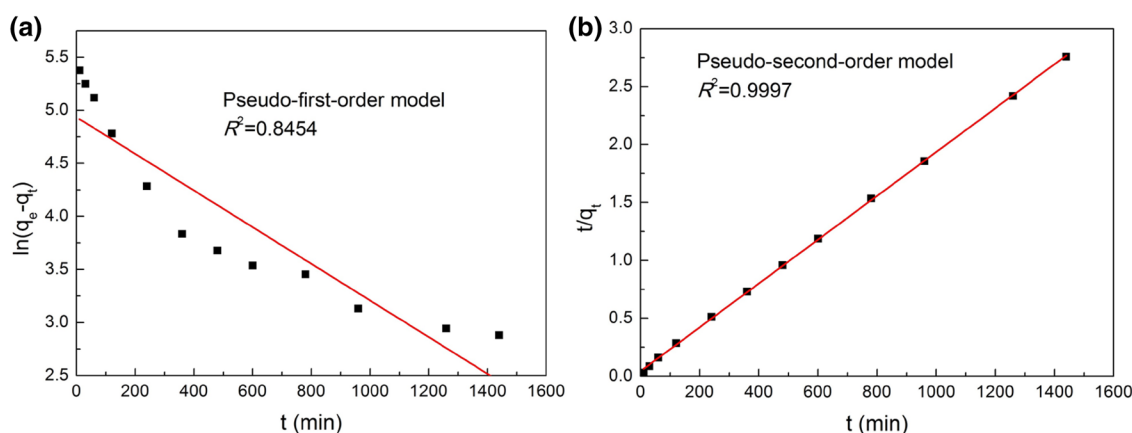


Fig. 10 The fitted plots of **a** pseudo-first-order model and **b** pseudo-second-order model for CIP adsorption onto MIL-101(Cr)-HSO₃

Table 1 Kinetic parameters of the adsorption of CIP onto MIL-101(Cr)-HSO₃

Sample	$q_{e,exp}$ (mg/g)	Pseudo-first-order model			Pseudo-second-order model		
		$q_{e,cal}$ (mg/g)	k_1 (min ⁻¹)	R^2	$q_{e,cal}$ (mg/g)	k_2 (g/(min mg))	R^2
MIL-101(Cr)-HSO ₃	522.2	139.0	1.73×10^{-3}	0.8454	529.1	8.087×10^{-5}	0.9997

model was 0.8454. Whereas the R^2 value for the pseudo-second-order model was 0.9997, which was much higher than that for the pseudo-first-order model. Additionally, the calculated adsorption capacity (529.1 mg/g) obtained by the pseudo-second-order model agreed better with the experimental value (522.2 mg/g). Hence, the adsorption behaviors of CIP adsorbed on MIL-101(Cr)-HSO₃ can be described well by the pseudo-second-order model. Moreover, it can be speculated that the adsorption process between CIP and MIL-101(Cr)-HSO₃ was attributed to the chemisorption process [65].

To find out the rate-limiting step of the adsorption process and verify the influence of mass transfer resistance on the CIP adsorption by MIL-101(Cr)-HSO₃, the adsorption kinetic data was fitted to the intra-particle diffusion model. The equation of intra-particle diffusion model was listed as follows.

$$\text{Intra-particle diffusion model: } q_t = k_d t^{0.5} + C \quad (4)$$

where t (min) is the adsorption time, q_t (mg/g) is the adsorption capacity at time t , k_d (g/(mg·min^{0.5})) is the intra-particle diffusion rate, and C (mg/g) is the thickness of boundary layer. Figure 11 displayed the fitted line of q_t versus $t^{0.5}$ and the parameters of intra-particle diffusion model were listed in Table 2. The results indicated that the adsorption process was controlled by more than one step. The large surface area and lots of adsorption sites on the adsorbent led to a rapid diffusion of CIP to the external surface of MIL-101(Cr)-HSO₃, which can be confirmed by the steep increase of the first stage. The second portion corresponded to the intraparticle diffusion and was ascribed to the final equilibrium stage, in which the intra-particle diffusion began to slow down due to the decreasing of the CIP concentration. It was evident that the fitted lines did not pass through

Table 2 Parameters of the intra-particle diffusion model

Sample	$q_{e,\text{exp}}$ (mg/g)	k_d (mg/(g·min ^{0.5}))	C (mg/g)	R^2
MIL-101(Cr)-HSO ₃	522.2	11.88	285.2	0.9945

the original point, suggesting that some other mechanisms may be involved in the CIP adsorption process along with the intra-particle diffusion [26, 66, 67]. Meanwhile, the intercept value of fitted line was relatively similar to the equilibrium adsorption capacity suggesting that film diffusion also played important role in CIP adsorption process by MIL-101(Cr)-HSO₃. Therefore, it can be deduced that both the intra-particle diffusion and film diffusion determined the adsorption process of MIL-101(Cr)-HSO₃ for CIP.

3.6 Adsorption isotherm

To determine the adsorption performance of CIP onto MIL-101(Cr)-HSO₃, the adsorption isotherm experiments were conducted at 303 K and the results were shown in Fig. 12. The isotherm was fitted by Langmuir, Freundlich, Tempkin and Dubinin–Radushkevich (D–R) models to investigate the adsorption mechanism. The equations of these models were expressed as follows:

$$\text{Langmuir model: } \frac{c_e}{q_e} = \frac{c_e}{q_m} + \frac{1}{q_m k_L} \quad (5)$$

$$\text{Freundlich model: } \ln q_e = \frac{1}{n} \ln c_e + \ln k_F \quad (6)$$

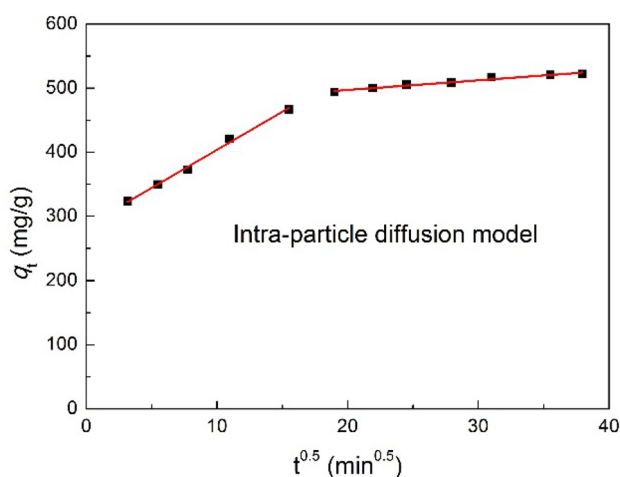


Fig. 11 The fitted plot of intra-particle diffusion model for CIP adsorption by MIL-101(Cr)-HSO₃

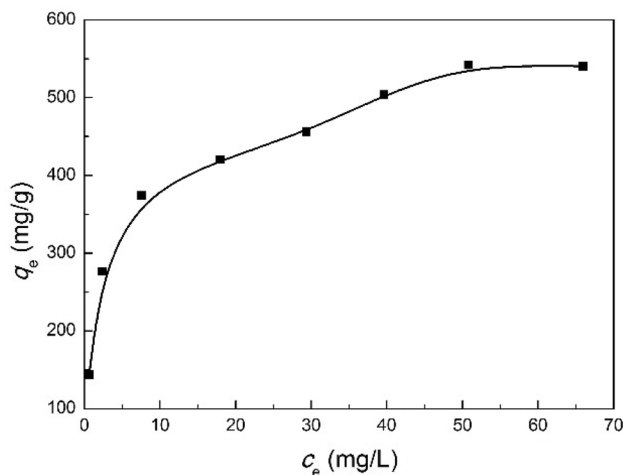


Fig. 12 Adsorption isotherm of CIP by MIL-101(Cr)-HSO₃ (adsorbent dosage = 0.1 g/L, T = 303 K, pH 8.0, contact time = 24 h)

$$\text{Tempkin model: } q_e = k_T \ln c_e + k_T \ln f \quad (7)$$

$$\text{Dubinin - Radushkevich model: } \ln q_e = \ln q_m - \beta \varepsilon^2 \quad (8)$$

where c_e (mg/L) is the equilibrium concentration of CIP, q_e (mg/g) is the equilibrium adsorption capacity and q_m (mg/g) is the calculated maximum adsorption capacity. k_L (L/mg) is the adsorption rate constant of Langmuir model, while k_F ((mg/g)(L/mg) $^{1/n}$) and $1/n$ are the adsorption constant and intensity constant of Freundlich model, respectively. k_T and f (L/mg) are the Tempkin constant and Tempkin binding constant, which reflect the adsorption heat and maximum combination energy, respectively. For D–R model, β (mol 2 /kJ 2) is the constant related to the mean free energy

of adsorption and ε is the Polanyi potential calculated by $\varepsilon = RT \ln(1 + 1/c_e)$, where R is 8.314 J/(mol K) and T (K) is the absolute temperature. The fitted linear lines of these isotherm models were shown in Fig. 13 and the adsorption model parameters were presented in Table 3. Obviously, the selected isotherm models were all suitable to fit the isotherm data because all of the correlation coefficient R^2 were higher than 0.94.

The Langmuir model suggests that the adsorption sites of adsorbent are covered by the adsorbate in the form of a monolayer, while the Freundlich model is based on multilayer adsorption. Obviously, the correlation coefficient of Langmuir model was 0.9926, which was the highest value among the obtained coefficients by the selected isotherm models.

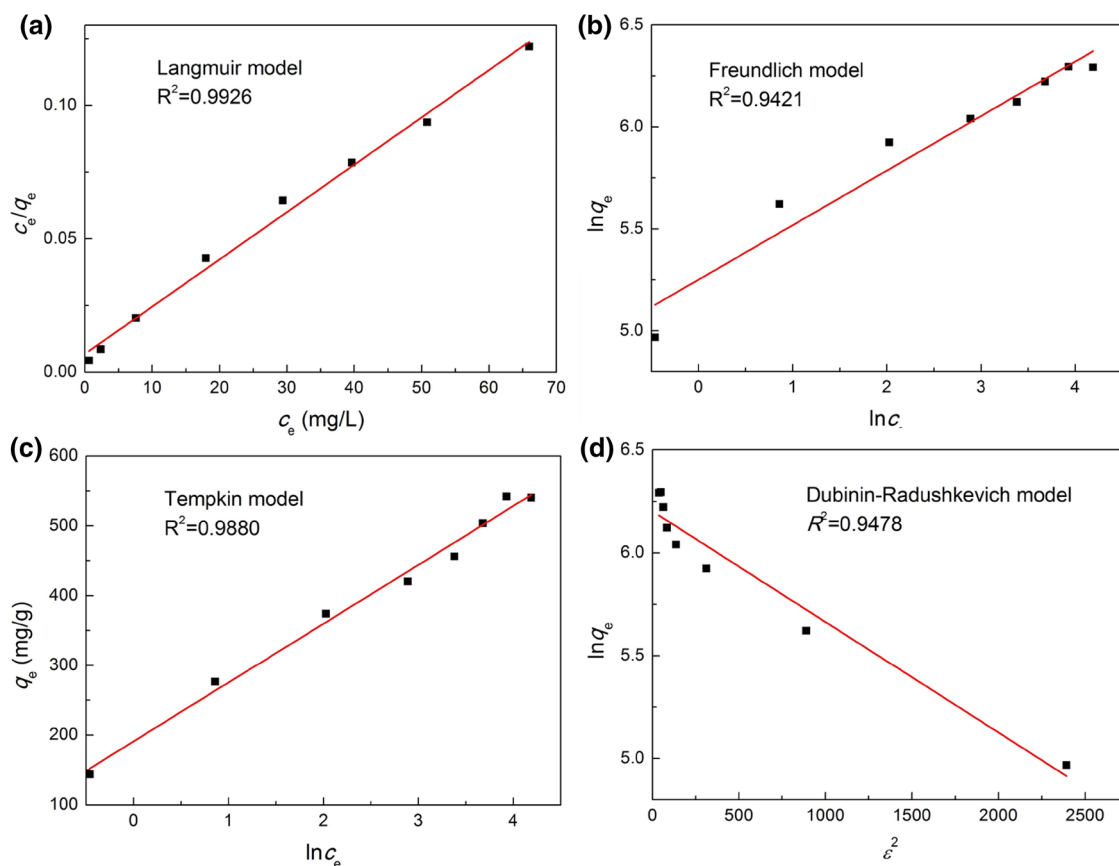


Fig. 13 Adsorption isotherms fitted by **a** Langmuir model, **b** Freundlich model, **c** Tempkin model and **d** Dubinin–Redushkevich model

Table 3 Isotherm parameters for the Langmuir, Freundlich, Tempkin and D–R models

$q_{e,iso}$ (mg/g)	Langmuir model			Freundlich model		
	q_m (mg/g)	k_L (L/mg)	R^2	k_F (mg/g)(L/mg) $^{1/n}$	$1/n$	R^2
540.5	564.9	0.2603	0.9926	190.69	0.2676	0.9421
$q_{e,iso}$ (mg/g)	Tempkin model			Dubinin–Radushkevich model		
	k_T (L/mg)	f	R^2	q_m (mg/g)	β (mol 2 /kJ 2)	R^2
540.5	84.34	9.645	0.9880	494.1	0.0005384	0.9478

This result indicated that most of the CIP molecules were adsorbed onto MIL-101(Cr)-HSO₃ by a mono-layer form. Whereas, the correlation coefficient of Freundlich model was also relatively higher (> 0.90), suggesting that multilayer adsorption also existed in the adsorption process of CIP onto MIL-101(Cr)-HSO₃. The similar result was also observed in the previous works [24]. What's more, the value of n was 3.74, falling between 2 and 10, which suggested that CIP can be adsorbed on MIL-101(Cr)-HSO₃ easily and the surface of adsorbent was heterogeneous [68]. A stronger electrostatic interaction between CIP and MIL-101(Cr)-HSO₃ can be suggested by the Tempkin model with $R^2 = 0.9880$, which was consistent with the conclusions obtained by the effect of pH [47]. As for the D-R model, E_a was usually used to determine the adsorption and its value can be calculated by $E_a = 1/(2\beta)^{0.5}$. Generally, $E_a < 8$ kJ/mol suggests that the adsorption process can be assigned to physical adsorption, while chemical adsorption, for example, ion exchange, is dominant with the value falling between 8 and 16 kJ/mol. If E_a ranging from 20 to 40 kJ/mol, the main adsorption mechanism can be described by chemisorption. For the CIP adsorption by MIL-101(Cr)-HSO₃, the value of E_a was 30.74 kJ/mol, indicating a chemisorption process. This result was consistent with the speculation obtained from the pseudo-second-order model. Based on the Langmuir model, the maximum adsorption capacity of MIL-101(Cr)-HSO₃ was 564.9 mg/g. To evaluate the CIP adsorption performance of MIL-101(Cr)-HSO₃, the maximum adsorption capacity of CIP was compared with that of the other materials, as listed in Table 4. As can be seen, the maximum adsorption capacity obtained with MIL-101(Cr)-HSO₃ was higher than that of other materials. Besides, the contact time was almost comparable with the other materials with relatively high adsorption capacity, such as graphene-soy protein aerogels [69], diethylenetriaminepentaacetic acid-functionalized magnetic graphene oxide [20], and ZIF-8 derived C@silica core/shell nanoparticles [7]. What's more, the adsorbent dosage was relatively lower than that of the most published materials. Compared with the MOF-derived porous carbon or composites, introducing -HSO₃ functional group on organic linker to enhance the adsorption performance of MIL-101(Cr) was simple and efficient. Therefore, MIL-101(Cr)-HSO₃ can be regarded as an efficient and promising adsorbent for the removal of CIP from aqueous solutions.

Temperature also affected the adsorption performance of CIP onto MIL-101(Cr)-HSO₃. The adsorption isotherms of CIP at different temperatures were depicted in Fig. 14. The equilibrium data obtained at 313 K and 323 K were also simulated by Langmuir and Freundlich model, respectively. The isothermal parameters and correlation coefficients (R^2) obtained from the selected models were listed on Table 5. It was evident that the Langmuir model described the sorption isotherms of CIP better than the Freundlich model under

different temperatures. At higher temperature, both mono-layer and multilayer adsorption also existed in the adsorption process of CIP onto MIL-101(Cr)-HSO₃, in which the former one dominated.

Besides, the effect of temperature on adsorption can be expressed by thermodynamic parameters including standard free energy change (ΔG°), standard enthalpy change (ΔH°), and standard entropy change (ΔS°). The values of ΔG° , ΔH° and ΔS° were calculated by the following equations:

$$\Delta G_o = -RT \ln K_o \quad (9)$$

$$\ln K_o = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \quad (10)$$

where R is an universal gas constant (8.314 J/mol K); T is the temperature in Kelvin. K_o is the thermodynamic equilibrium constant, obtained by plotting $\ln(q_e/c_e)$ versus c_e and extrapolating c_e to zero. The value of ΔG° was calculated with Eq. (9), while the values of ΔH° and ΔS° were obtained from the slope and intercept of the plot of $\ln K_o$ versus $1/T$ using the Van't Hoff Eq. (10). The calculated thermodynamic parameters were listed in Table 6. The negative values of ΔG° at all temperatures studied indicated that the adsorption of CIP onto MIL-101(Cr)-HSO₃ was feasible and spontaneous. What's more, the absolute value of ΔG° decreased with increasing temperature illustrating that the adsorption of CIP was exothermic and favored at lower temperature. The negative value of ΔH° also confirmed the exothermic adsorption process of CIP. The negative value of ΔS° suggested that the entropy decreased and also revealed the decreased randomness at solid-liquid interface during the adsorption process [20, 23, 26].

3.7 Regeneration of MIL-101(Cr)-HSO₃

Reusability is an important parameter for commercial applications. It indicates the cost effectiveness of an adsorbent. To investigate the reusability of MIL-101(Cr)-HSO₃, the MIL-101(Cr)-HSO₃ adsorbed CIP was added to 250 mL 0.1 ml/L HCl solution and the flask was put in a shaker at 313 K for 24 h. Then, the regenerated MIL-101(Cr)-HSO₃ was obtained by filtration and dried at 353 K for 12 h. The stability of the MIL-101(Cr)-HSO₃ was confirmed by XRD. The XRD patterns of the MIL-101(Cr)-HSO₃ adsorbed CIP and regenerated MIL-101(Cr)-HSO₃ were illustrated in Fig. S5. Obviously, the XRD of the two materials did not change, suggesting that MIL-101(Cr)-HSO₃ kept stable during the adsorption process and regenerated MIL-101(Cr)-HSO₃ also kept stable for several runs' adsorption-desorption. Besides, the TEM results shown in Fig. S6 indicated that both the pristine MIL-101(Cr)-HSO₃ and regenerated MIL-101(Cr)-HSO₃ had slight aggregation and they had octahedral

Table 4 Comparison of adsorption capacities and other parameters for the removal of CIP with other materials

Adsorbent	pH	Temperature (K)	Dosage (g/L)	Equilibrium time (h)	$q_{\max}$ (mg/g)	References
ZIF-8 derived C@silica core/shell nanoparticles	6.0	303	0.05	20	516.8	[7]
<i>Arundo donax</i> Linn AC ^a	5.05	298	1.0	24	244	[19]
Pomelo peel AC ^a	5.05	298	1.0	24	400	
DDMGO	8.0	303	0.09	24	230	[20]
Fe ₃ O ₄ @MIL-100(Fe)	6.0	298	0.50	5	322.58	[23]
Fe ₃ O ₄ /Cu-BTC	7.0	298	0.04	0.5	538	[24]
<i>Enteromorpha prolifera</i> derived AC ^a	9.0	303	1.2	36	233–286	[25]
AC ^a	7.92	298	0.40	5	454.68	[26]
Montmorillonite	6.0	293	0.25	4	330	[27]
Multi-walled CNTs ^b	4.0	298	0.5	24	192.4	[29]
Graphene oxide monolayers nanosheets	6.30	298	5.0	–	282	[30]
Fe-MCM-41	5.4	303	0.2	2	93.79	[33]
MIL-101(Cr)/Fe ₃ O ₄	8.0	298	–	0.83	63.28	[45]
ZIF-8 derived nanoporous carbon	6.0	303	0.04	18	416.7	[46]
ZIF-67 derived hollow Co ₃ S ₄	7.0	303	0.04	6	471.7	[47]
Ordered mesoporous carbon	6.0	298	2.0	24	267.9	[67]
Graphene-soy protein aerogels	4.0	298	0.5	24	500	[69]
Graphene oxide	5.0	–	–	48	379	[70]
CMK-1 bamboo-based carbon	6.0	298	1.0	24	233.4 362.9	[71]
KMS-1	4.0	313	0.133	7	269.5	[72]
MCGO ^c	5.0	–	0.33	8	282.9	[73]
Ordered mesoporous carbon	6.0	298	0.60	24	236	[74]
Carbon nanofibers	7.0	298	0.25	8	225.35	[75]
Ethylene diaminetetraacetic acid- β -cyclodextrin	4.0	298	0.3	4	327.1	[76]
Graphene hydrogel	–	298	–	120	312	[77]
Magnetic multifunctional resins	5.6	293	0.1	2	322.7	[78]
Porous graphene hydrogel	8.0	298	–	120	235.6	[79]
MIL-101(Cr)	–	303	0.1	24	113.2	This work
MIL-101(Cr)-HSO ₃	8.0	303	0.1	24	564.9	This work

^aActivated carbon^bShort for carbon nanotubes^cShort for Magnetic chitosan grafted graphene oxide nano-composite

microcrystals with sizes of the range 200–500 nm, which was consistent with the SEM results. The pore sizes of pristine MIL-101(Cr)-HSO₃ and regenerated MIL-101(Cr)-HSO₃ were around 3.85 nm. The results also suggested the stability of MIL-101(Cr)-HSO₃ during the adsorption and desorption process.

The adsorption capacity of the regenerated adsorbents was measured under the optimized conditions and the results were provided in Fig. S7. As can be seen from Fig. S7, the adsorption capacity of the regenerated MIL-101(Cr)-HSO₃ was around 296 mg/g, which was much lower than that of the raw MIL-101(Cr)-HSO₃. The results suggested that MIL-101(Cr)-HSO₃ was difficult to be regenerated, which could be assigned to the strong interactions between the

open Cr(III) sites in MIL-101(Cr)-HSO₃ and COO[−] groups in CIP.

3.8 Adsorption mechanism

In order to further investigate the commercial application of adsorbents, it is very important to clarify the adsorption mechanism of an adsorbent, which will be helpful for the further exploration and design of new adsorbents. Based on the experimental results discussed above, it can be concluded that the mechanism of adsorption process of CIP adsorbed on MIL-101(Cr)-HSO₃ are attributed to a combination of the electrostatic interactions, π – π interactions and the formation of coordination bonds.

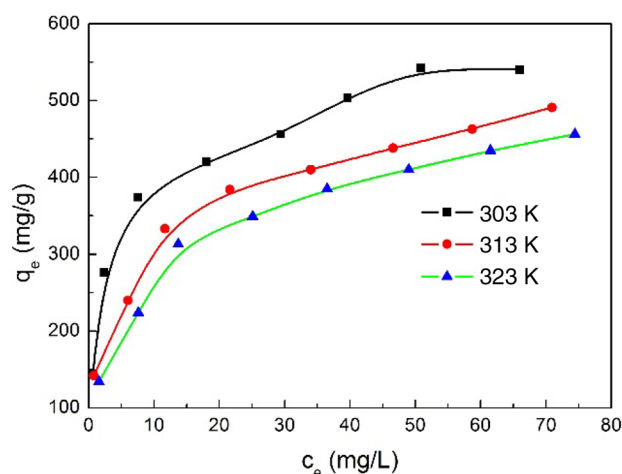


Fig. 14 Adsorption isotherms of CIP by MIL-101(Cr)-HSO₃ at different temperatures (adsorbent dosage = 0.1 g/L, pH 8.0, contact time = 24 h)

The adsorption mechanism was schematically shown in Fig. 15. Remarkably, the optimal value of pH was 8.0 and the most of the CIP existed as zwitterion molecules, which suggested that hydrophobic interactions can be regarded as an important role affecting the adsorption process of MIL-101(Cr)-HSO₃ [24, 68]. Moreover, the π - π interactions between the phenyl rings of MIL-101(Cr)-HSO₃ and CIP and the coordination bonds between the open Cr(III) sites in MIL-101(Cr)-HSO₃ and COO⁻ groups in CIP also played important roles in the adsorption behaviors. XPS measurements were carried out to validate the coordination bonds between the open Cr(III) sites in MIL-101(Cr)-HSO₃ and COO⁻ groups in CIP. The XPS spectra of Cr 2p spectrum of MIL-101(Cr)-HSO₃ and MIL-101(Cr)-HSO₃ adsorbed CIP were shown in Fig. S8. The peaks at the binding energy of around 587.3 eV showed a slight shift to the lower binding energy in the spectra of MIL-101(Cr)-HSO₃ adsorbed CIP. This slight shift of Cr 2p binding energy may arise from the negatively charged COO⁻ groups of CIP interacted with the open metal Cr(III) sites of MIL-101(Cr)-HSO₃, which increased the electron density around Cr(III) resulting in lowering the binding energy. This slight shift of Cr 2p can validate the strong interaction between the COO⁻ groups of CIP

with the open metal Cr(III) sites of MIL-101(Cr)-HSO₃ [80]. To investigate the π - π interactions between the phenyl rings of MIL-101(Cr)-HSO₃ and CIP, the adsorption location of CIP in the pores of MIL-101(Cr)-HSO₃ was explored through molecular simulation. During the simulation process, the charges of CIP molecules and MIL-101(Cr)-HSO₃ were set as zero to avoid the effect of the electrostatic interactions. The simulation results with the lowest energy were shown in Fig. S9. As shown in Fig. S9, we could clearly observe that the characteristic distances between the benzene-ring centers of one CIP molecule and the organic linkers of MIL-101(Cr)-HSO₃ were 4.377 Å, which was indicative of forming the weak π - π interactions [81]. Based on the zeta potentials of MIL-101(Cr)-HSO₃ and the formation species of CIP at different pH, the electrostatic interactions also affected the adsorption process, which can be proved by the decrease of the adsorption capacity of MIL-101(Cr)-HSO₃ at higher concentration of NaCl in the solution. The obvious decrease was caused by that the surface charge of MIL-101(Cr)-HSO₃ was neutralized by the excess dissociative-NaCl surrounding the adsorption sites, hindering the electrostatic interactions. Also, the positively charged groups of the zwitterionic CIP molecules were attracted to the negatively charged surface of MIL-101(Cr)-HSO₃ whilst the negatively charged groups of CIP were coordinated to the open Cr(III) sites of MIL-101(Cr)-HSO₃, which may result in a more favorable molecular alignment in the pores of MIL-101(Cr)-HSO₃ and higher binding affinity and adsorption capacity [64]. Generally speaking, the interactions described above promote the CIP adsorption performance of MIL-101(Cr)-HSO₃ together.

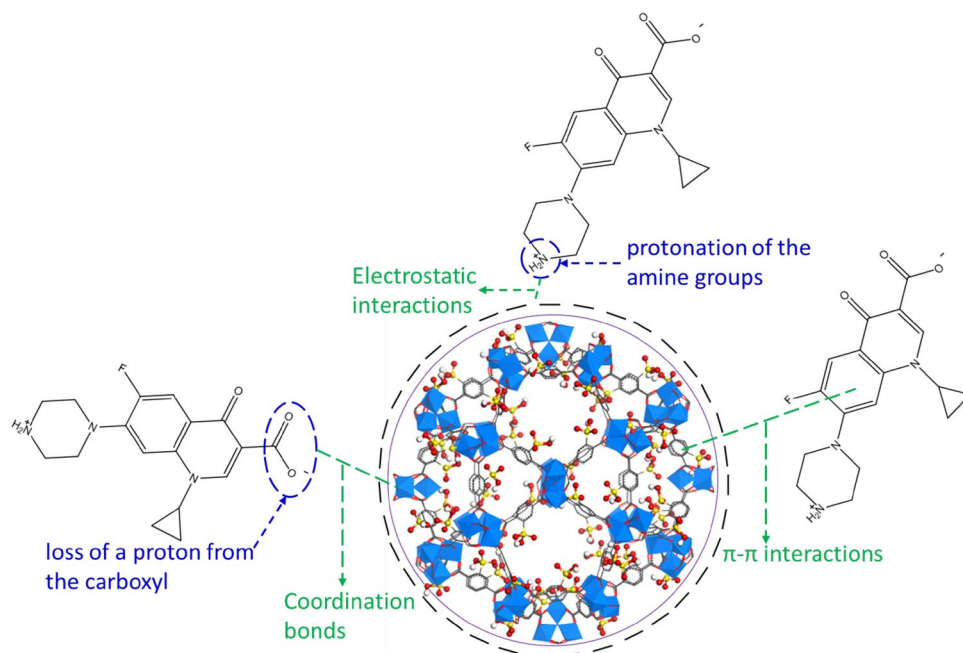
Table 6 Adsorption thermodynamic parameters of CIP on MIL-101(Cr)-HSO₃

Temperature (K)	$\ln K_0$	ΔG (kJ/mol)	ΔH (kJ/mol)	ΔS (J/K mol)
303	4.591	-11.564	-33.818	-73.518
313	4.135	-10.760	-	-
323	3.760	-10.092	-	-

Table 5 Parameters of adsorption isotherm of CIP on MIL-101(Cr)-HSO₃ at different temperatures

Temperature (K)	Langmuir model			Freundlich model		
	q_m (mg/g)	k_L (L/mg)	R^2	k_F (mg/g) (L/mg) ^{1/n}	$1/n$	R^2
303	564.9	0.2603	0.9926	190.69	0.2676	0.9421
313	510.2	0.1687	0.9910	153.95	0.2778	0.9840
323	490.2	0.1258	0.9922	121.35	0.3173	0.9798

Fig. 15 Possible adsorption mechanism of CIP on MIL-101(Cr)-HSO₃ at pH 8.0



4 Conclusions

In this work, the adsorption performance of CIP on MIL-101(Cr)-HSO₃ was investigated. The optimal adsorption conditions were determined, such as adsorbent dosage, pH, salt concentration, and contact time. Based on the experimental data, the maximum adsorption capacity of CIP calculated by Langmuir model was 564.9 mg/g, which was higher than most of the other porous materials reported in the previous works. The adsorption kinetic study showed that the adsorption equilibrium reached within 24 h, which was comparable to that of the previously reported adsorbents. Considering the pH dependent adsorption capacity of CIP and the zeta potentials of the adsorbents, it can be concluded that the electrostatic interactions, hydrophobic interactions, and π - π interactions played important roles in the CIP adsorption by MIL-101(Cr)-HSO₃ together. Our results demonstrated that MIL-101(Cr)-HSO₃ with high porosity and negative surface charges can be considered as a promising candidate to remove CIP from aqueous solutions. By the way, adding functional groups on MOFs may be an efficient way to change the surface charges, then enhancing the adsorption performance of MOFs for the pH dependent water pollutants.

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